

1. General Description

The EMS7227 is a bi-directional, low-power, two-port, High-Speed, USB2.0 switch. Configured as a doublepole, double-throw switch (DPDT) switch, it is optimized for switching between any combination of High-Speed (480Mbps) or Full-Speed (12Mbps) source.

The EMS7227 is compatible with the requirements of USB2.0 and features an extremely low on capacitance (CON) of 3.7pF. The wide bandwidth of this device (720MHz) exceeds the bandwidth needed to pass the third harmonic, resulting in signals with minimum edge and phase distortion. Superior channel-to-channel crosstalk also minimizes interference.

The EMS7227 contains special circuitry on the switch I/O pins for applications where the V_{CC} supply is powered-off ($V_{CC} = 0$), which allows the device to withstand an overvoltage condition. This device is designed to minimize current consumption even when the control voltage applied to the S pin is lower than the supply voltage (V_{CC}). This feature is especially valuable to ultra-portable applications, such as cell phones, allowing for direct interface with the general purpose I/Os of the baseband processor. Other applications include switching and connector sharing in portable cell phones, PDAs, digital cameras, printers, and notebook computers.

2. Features and Benefits

- Low On Capacitance: 3.7pF Typical
- Low On Resistance: 3.0Ω Typical
- Low Power Consumption: $1\mu A$ Maximum - $15\mu A$ Maximum I_{CCT} over an Expanded Voltage Range ($V_{IN} = 1.8V, V_{CC} = 4.4V$)
- Wide -3db Bandwidth: $> 720MHz$
- 8kV ESD Rating, $> 16kV$ Power/GND ESD Rating
- Over-Voltage Tolerance (OVT) on all USB Ports Up to 5.25V without External Components

3. Applications

- Cell phone, PDA, Digital Camera, and Notebook
- LCD Monitor, TV, and Set-Top Box

4. Ordering Information

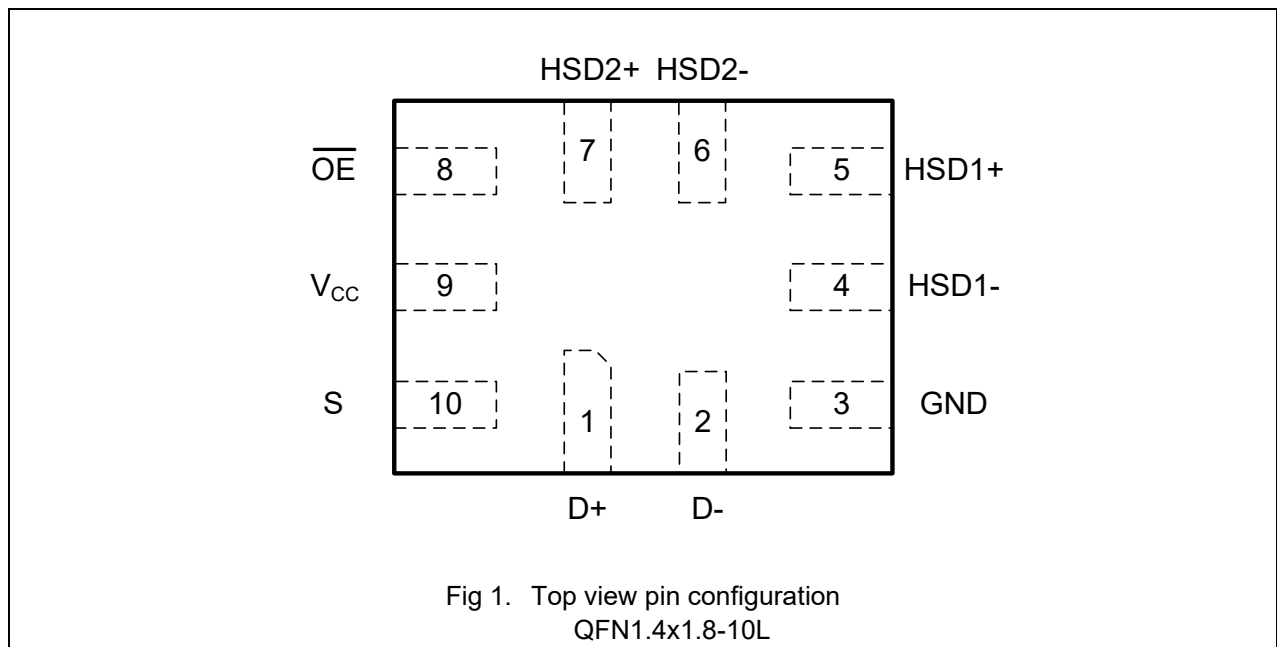
Table 1. Ordering information

Type number	Topside marking [1]	Package		Quantity
		Name	Description	
EMS7227RSW	7227	QFN1.4×1.8-10L	QFN package, 10 pins 1.4 mm × 1.8 mm; 0.5 mm (Max) height	4000
EMS7227MS	7227	MSOP3.0×3.0-10L	MSOP package, 10 pins 3.0 mm × 3.0 mm; 1.1 mm (Max) height	4000

[1] There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

5. Pinning Information

5.1. Pin map



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Low-Power, Two-Port, Hi-Speed, USB2.0 (480Mbps) UART Switch

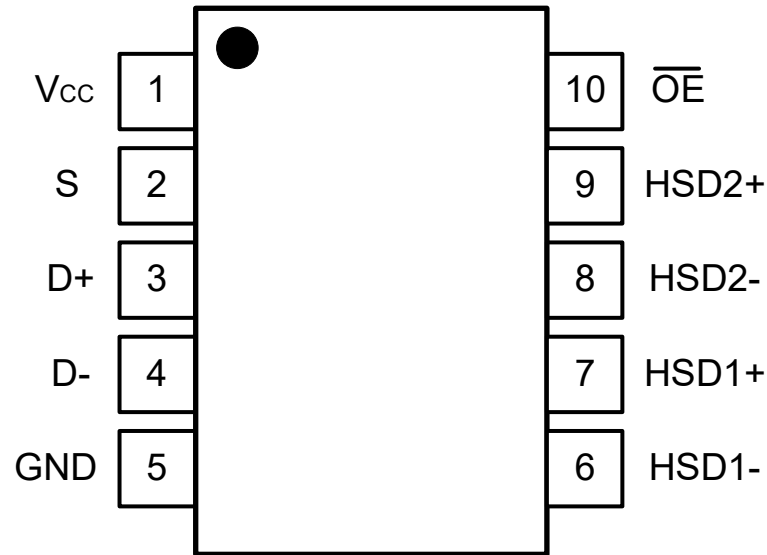


Fig 2. Top view pin configuration
MSOP3.0x3.0-10L

5.2. Pin description

Table 2. Pin description

Symbol	Package		Description
	QFN-10L	MSOP-10L	
D+	1	3	Common USB Data Bus
D-	2	4	Common USB Data Bus
GND	3	5	Ground
HSD1-	4	6	Multiplexed Source Input 1
HSD1+	5	7	Multiplexed Source Input 1
HSD2-	6	8	Multiplexed Source Input 2
HSD2+	7	9	Multiplexed Source Input 2
$\overline{OE}$	8	10	Switch Enable (HIGH: DISABLE; LOW: ENABLE)
V _{CC}	9	1	Supply Voltage
S	10	2	Switch Select (LOW: D+, D- = HSD1+, HSD1-; HIGH: D+, D- = HSD2+, HSD2-)

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

Input S	Switch
L	HSD1+ to D+, HSD1- to D-
H	HSD2+ to D+, HSD2- to D-

7. Typical Application

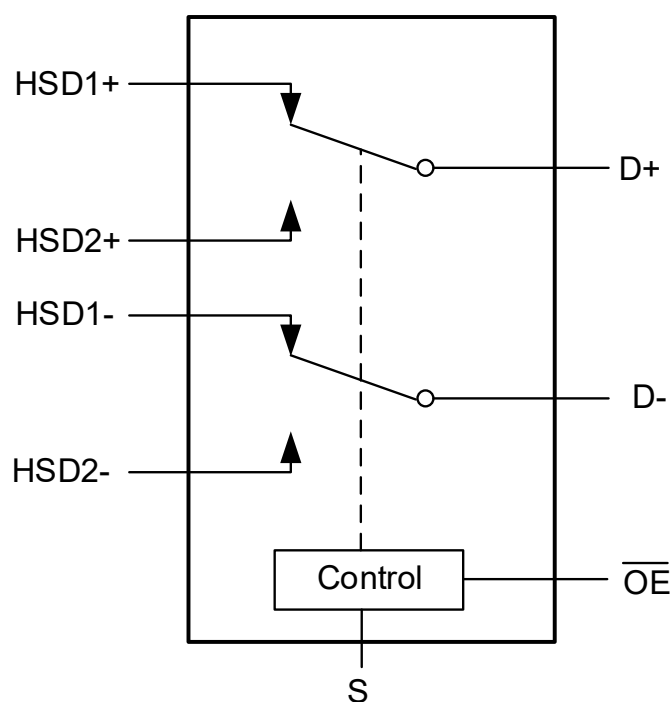


Fig 3. Typical Application Circuit

8. Absolute Maximum Ratings

Absolute Maximum Ratings may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification is not implied.

Table 4. Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	Supply voltage		-0.5	5.6	V
V _{SW}	DC switch I/O voltage (V _{CC} = 0V)		-0.5	5.25	V
V _{CNTRL}	DC input voltage (S, $\overline{OE}$) [1]		-0.5	V _{CC}	V
I _{IK}	DC input diode current		-50		mA
I _{out}	DC output current			100	mA
T _{stg}	Storage temperature		-65	150	°C
MSL	Moisture Sensitivity Level (JEDEC J-STD-020A)			1	Level
ESD	Human Body Model, JEDEC: JESD22-A114	All Pins	7		kV
		I/O to GND	8		
		Power to GND	16		
		D+/D-	9		
	IEC 61000-4-2 System on USB Connector Pins D+ & D-	Air Discharge	15		
		Contact	8		
	Charged Device Model, JEDEC: JESD22-C101		2		

[1] The input and output negative ratings may be exceeded if the input and output diode current ratings are observed.

9. Recommended Operating Conditions

Table 5. Recommend operating ratings

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	Supply voltage operating		2.4	4.4	V
V _{CNTRL}	Control input voltage (S, $\overline{OE}$) [1]		0	V _{CC}	V
V _{SW}	Switch I/O voltage		-0.5	4.5	V
T _A	Operating temperature		-40	85	°C

[1] The control input must be held HIGH or LOW and it must not float.

10. Electrical Characteristics

Table 6. Static characteristics

All typical values are for $V_{CC} = 3.3V$ at $T_A = 25^{\circ}C$ unless otherwise specified.

Symbol	Parameter	Conditions	T _A = -40°C to +85°C			Unit
			Min	Typ	Max	
DC Characteristics						
V _{IK}	Clamp diode voltage	V _{CC} = 3.0 V, I _{IN} = -18 mA			-1.2	V
V _{IH}	Input logic high level	V _{CC} = 2.4 V to 3.6 V	1.3			V
		V _{CC} = 4.3 V	1.7			V
V _{IL}	Input logic low level	V _{CC} = 2.4 V to 3.6 V			0.5	V
		V _{CC} = 4.3 V			0.7	V
I _{IN}	Control input leakage	V _{SW} = 0 to V _{CC} , V _{CC} = 0 to 4.3 V	-1.0		1.0	μA
I _{OZ}	Off state leakage	V _{CC} = 4.3 V, 0 ≤ D _n , HSD1n, HSD2n ≤ 3.6 V	-2.0		2.0	μA
I _{OFF}	Power-off leakage current (All I/O Ports)	V _{CC} = 0 V, V _{SW} = 0 V to 4.3 V	-2.0		2.0	μA
R _{ON}	HS switch on resistance [1]	V _{CC} = 2.4 V, V _{SW} = 0.4 V, I _{ON} = -8 mA		4.5	7.5	Ω
		V _{CC} = 3.0 V, V _{SW} = 0.4 V, I _{ON} = -8 mA		3.5	6.5	Ω
ΔR _{ON}	HS delta RON [2]	V _{CC} = 3.0 V, V _{SW} = 0.4 V, I _{ON} = -8 mA		0.65		Ω
I _{CC}	Quiescent supply current	V _{CC} = 4.3 V, V _{CNTRL} = 0 or V _{CC} , I _{OUT} = 0			1	μA
I _{CCT}	Increase in I _{CC} current per control voltage and V _{CC}	V _{CC} = 4.3 V, V _{CNTRL} = 2.6 V			10	μA
		V _{CC} = 4.3 V, V _{CNTRL} = 1.8 V			15	μA
AC Characteristics						
C _{IN}	Control pin input capacitance	V _{CC} = 0 V		1.5		pF
C _{ON}	D+/D- on capacitance	V _{CC} = 3.3 V, $\overline{OE}$ = 0 V, f = 240 MHz		3.7		pF
C _{OFF}	D1n, D2n off capacitance	V _{CC} and $\overline{OE}$ = 3.3 V		2.0		pF
t _{ON}	Turn-on time S, $\overline{OE}$ to output	V _{CC} = 2.4 V, V _{SW} = 0.8 V, C _L = 5 pF, R _L = 50 Ω		24	40	ns
		V _{CC} = 3.0 to 3.6 V, V _{SW} = 0.8 V, C _L = 5 pF, R _L = 50 Ω		13	30	ns
t _{OFF}	Turn-off time S, $\overline{OE}$ to output	V _{CC} = 2.4 V, V _{SW} = 0.8 V, C _L = 5 pF, R _L = 50 Ω		15	25	ns
		V _{CC} = 3.0 to 3.6V, V _{SW} =0.8V,C _L =5pF,R _L =50Ω		12	35	ns

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t_{PD}	Propagation delay [3]	$V_{CC} = 3.3 \text{ V}, C_L = 5 \text{ pF}, R_L = 50 \Omega$		0.25		ns
t_{BBM}	Break-before-make time	$V_{CC} = 2.4 \text{ V}, V_{SW1} = V_{SW2} = 0.8 \text{ V}, C_L = 5 \text{ pF}, R_L = 50 \Omega$	2.0		10	ns
		$V_{CC} = 3.0 \text{ to } 3.6 \text{ V}, V_{SW1} = V_{SW2} = 0.8 \text{ V}, C_L = 5 \text{ pF}, R_L = 50 \Omega$	2.0		6.5	ns
O_{IRR}	Off isolation	$V_{CC} = 3.0 \text{ to } 3.6 \text{ V}, R_L = 50 \Omega, f = 240 \text{ MHz}$		-30		dB
X_{TALK}	Non-Adjacent channel crosstalk	$V_{CC} = 3.0 \text{ to } 3.6 \text{ V}, R_L = 50 \Omega, f = 240 \text{ MHz}$		-45		dB
BW	-3dB bandwidth	$V_{CC} = 3.0 \text{ to } 3.6 \text{ V}, R_L = 50 \Omega, C_L = 0 \text{ pF}$		720		MHz
		$V_{CC} = 3.0 \text{ to } 3.6 \text{ V}, R_L = 50 \Omega, C_L = 5 \text{ pF}$		550		MHz
$t_{SK(P)}$	Skew of opposite transitions of the same output	$V_{CC} = 3.3 \text{ V}, R_L = 50 \Omega, C_L = 5 \text{ pF}$		20		ps
T_J	Total jitter [4]	$V_{CC} = 3.3 \text{ V}, R_L = 50 \Omega, C_L = 5 \text{ pF}, t_R=t_F=500 \text{ ps}(10\text{-}90\%) \text{ at } 480 \text{ Mbps (PRBS} = 2^{15}\text{-}1)$		200		ps

[1] Measured by the voltage drop between HSDn and Dn pins at the indicated current through the switch. On resistance is determined by the lower of the voltage on the two (HSDn or Dn ports).

[2] [3] [4] Guaranteed by characterization.

11. Test Circuits

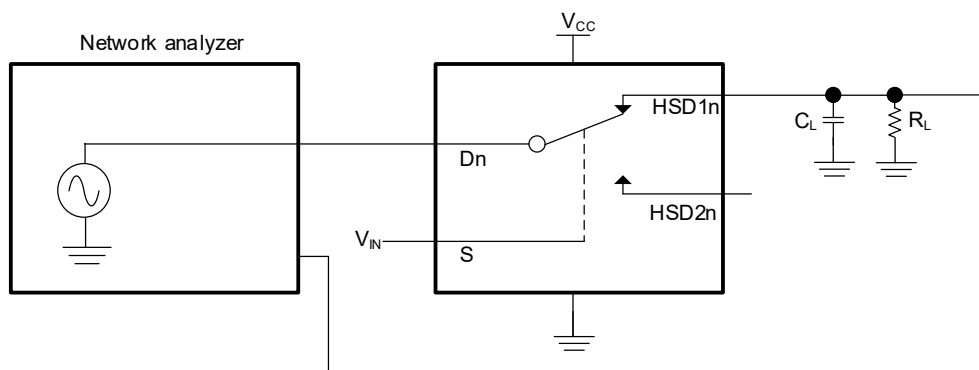


Fig 4. Bandwidth

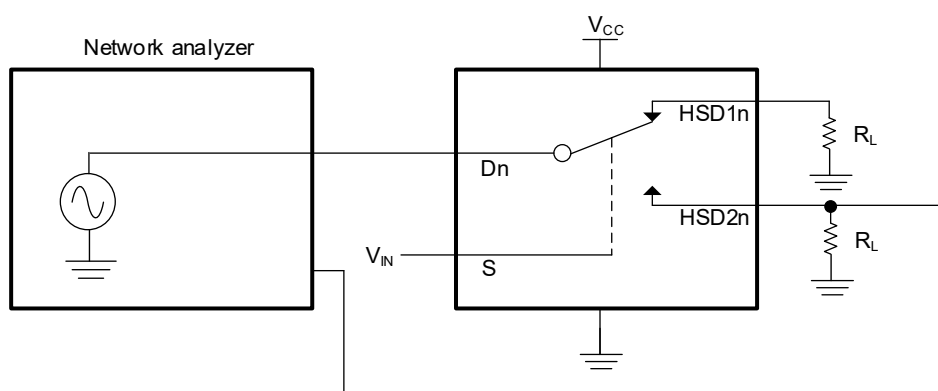


Fig 5. Off isolation

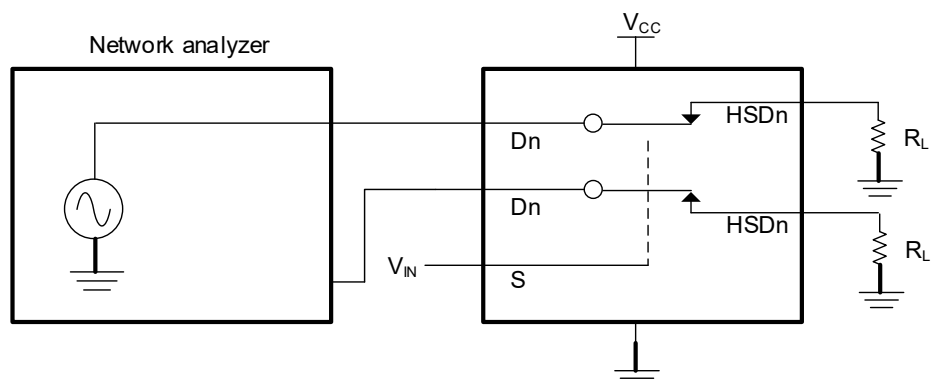
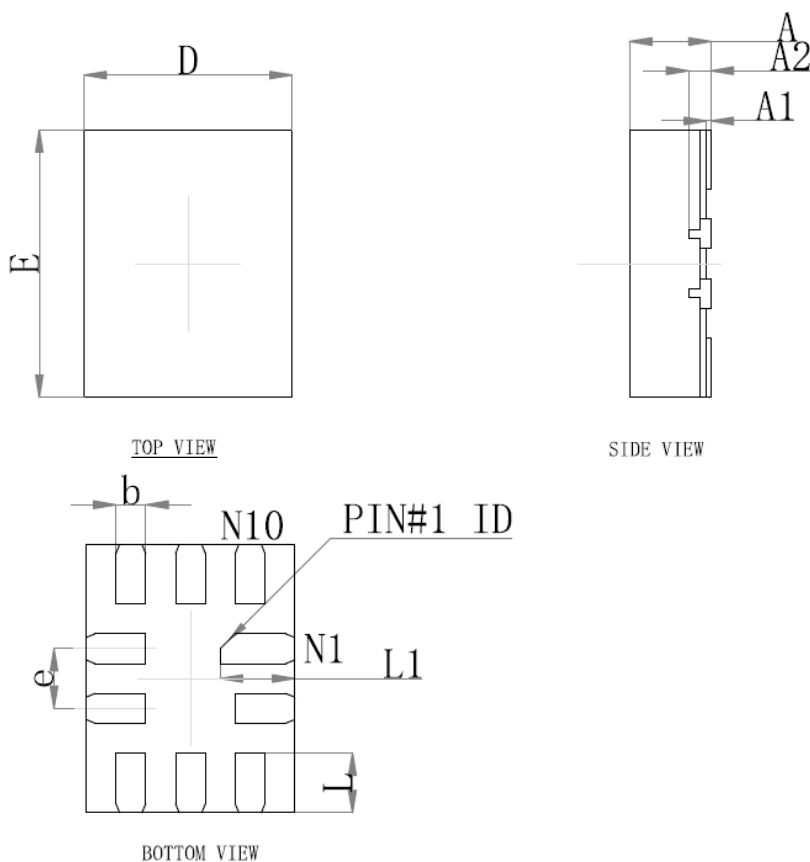


Fig 6. Crosstalk

12. Package Outline

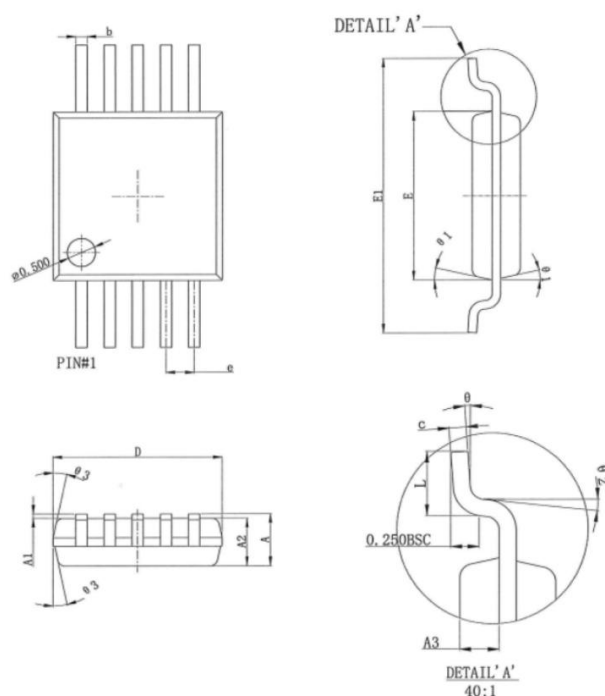
QFN1.4x1.8 -10L


Symbol	Dimension in Millimeters		
	Min.	NOM.	Max.
A	0.400	0.450	0.500
A1	0.000	0.020	0.050
A2	0.127 Ref.		
D	1.300	1.400	1.500
E	1.700	1.800	1.900
b	0.150	0.200	0.250
e	0.400 Typ.		
L	0.350	0.400	0.450
L1	0.450	0.500	0.550

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MSOP3.0×3.0-10L



Symbol	Dimension in Millimeters		
	Min.	NOM.	Max.
A	0.820	0.930	1.100
A1	0.020	0.080	0.150
A2	0.750	0.850	0.950
A3	0.249	0.349	0.449
c	0.090	0.152	0.230
E	2.900	3.000	3.100
E1	4.750	4.900	5.050
L	0.400	0.600	0.800
b	0.180	0.200	0.280
D	2.900	3.000	3.100
e	0.400 Typ.		
θ	0°	3°	6°
θ1	12°Ref.		
θ2	4°Ref.		
θ3	12°Ref.		

13. Revision History

Table 7. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS7227 Rev.0.1	Oct 09, 2025	Draft datasheet		